GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY
Dept. of Science and Technology, Govt. of Gujarat

ROBOFEST- GUJARAT 6.0

Guidelines for submission of Ideation Proposal
Senior category

All teams must be present and ready to perform their challenge. The committee may assign any team to the task at random. Any kind of absence and lack of readiness will be subject to suitable actions, including the deduction of score/disqualification.

Students are required to perform their task and answer the questions. Mentors should not be involved in the task. The committee may allow mentors during question-answer if it feels necessary, depending on the situation.

Sample images given in these guidelines are only for visual understanding.

Autonomous Underwater Vehicle

Game Description:

The AUV Challenge aims to simulate real-world underwater navigation, detection, and manipulation tasks using an Autonomous Underwater Vehicle (AUV). The competing AUV must autonomously navigate an obstacle-free underwater arena, identify specific visual and acoustic targets, and successfully drop a payload into designated target drums—without surfacing, touching the pool walls or floor, or requiring remote control.

This challenge will test participants' skills in underwater robotics, sensor fusion, path planning, acoustic signal processing, and target acquisition using onboard computing.

Challenge Tasks:

1. Qualification Gate Run:
    - The AUV must pass through a gate forward, perform a U-turn, and return through the gate in reverse.
    - Points are awarded only when forward and reverse runs are completed without touching pool sides/bottom.
2. Visual Target Zone Navigation:
    - Navigate to the green mat zone (8m x 2m) placed on the pool floor using computer vision.
    - Identify one or multiple submerged objects via QR code scan without stopping the motion of the AUV
3. Target Drum Identification:
    - Identify and localize one blue drum among three red drums.
    - One red drum will emit an acoustic ping (using a pinger of 40 kHz), randomly placed at each attempt.
4. Target Acquisition:
    - Drop a payload (medium -sized ball:diameter: ~70 mm) into the designated drum:
        - Blue drum: 30 points
        - Pinger drum: 20 points
        - Other red drum: 10 points

      Material: Plastic. Weight: 2 kg (max.)
5. Bonus Challenge:
    - Generate and display a digital 2D map of the mat and drum positions (to be reviewed post-mission).

Game Field Design:

- Arena: Rectangular water tank/pool (Minimum size: 10m x 6m, depth: 1.5m+).

- Start Zone: Marked start line near one short side of the pool.
- Qualification Gate: Placed 2 meters ahead of the start line. Approximate dimensions of the Qualification gate: 5 ft x 5 ft. (Length X Height). The approximate width of the gate will be 4-5 inches. The gate will be positioned below the water surface. Exact locations of the qualification starting line and gate within the pool can't be provided a-priori.
- Green Target Zone: Located on the opposite half of the pool, four colored drums are placed in random order on a green mat.
- Drums: Equally spaced within the green zone. Each drum is identifiable by its color.
- Obstacle-Free: No physical obstacles or barriers in the pool; emphasis is on autonomy and accuracy.
- Size & weight of the AUV will depend on the specific design. There is no prescribed limit on dimensions & weight. There is no restriction on external appendages.
- Time will be measured by GUJCOST authority using usual means, e.g. stop watch. The run-time will be used primarily for scoring. Ideally, the AUV has to complete the run within 10 minutes time limit. Nonetheless, the decision for allowing a team to run the AUV for longer duration will be made jointly by the TAC & GUJCOST officials, based on the technical merit of the hardware.
- There is no energy limit for the battery. Team can select suitable battery-pack based on the design. There is no restriction on the type of battery. However, a kill-switch reachable from the poolside is mandatory.
- Approximate dimensions of the drums: circular drums with 2 ft diameter and 4 ft height. Drums will be made of suitable non-metallic materials. Frequency and intensity / specs of acoustic pinger and duration of each ping can be provided to the teams in due course, on request.

Design Specialties Expected:

- Modular watertight chassis
- Reliable battery system for 10+ minutes of operation
- Vision systems (camera-based) for drum identification
- Hydrophone/Acoustic sensor for pinger localization
- Controlled buoyancy and stabilization
- Robotic arm or drop mechanism through suitably-designed gripper for payload release

Stage 1: Proof of Concept (PoC)

Objective

To evaluate the **fundamental capabilities** of the AUV system including:

- Stability and buoyancy control
- Basic navigation
- Sensor integration (vision/acoustic optional)
- Payload handling mechanism

PoC Challenge Tasks

1. Basic Navigation Test

- AUV must:
    - Move forward for a defined distance
    - Execute a controlled turn
    - Return to start point

2. Qualification Gate Run (Simplified)

- Pass through a gate **once (forward only)**
- No contact with pool walls or gate

3. Visual Detection Task

- Identify **one coloured object (drum/marker)**
- Scan the QR code of at least one object and identify
- Position estimation not mandatory

4. Basic Payload Drop

- Drop payload in a **designated open zone**
- Precision not strictly evaluated

5. Stability & Control Demonstration

- Maintain depth and orientation for 30–60 seconds

Technical Constraints (PoC Stage)

- Fully autonomous OR semi-autonomous allowed
- Limited external intervention allowed (safety only)
- No restriction on size/weight (as per 5.0 guidelines)

Evaluation Criteria (PoC)

Criteria	Marks
Navigation Control	20
Stability & Buoyancy	20
Gate Navigation	15
Object Detection	15
Payload Mechanism	10
System Integration	10
Innovation	10

Stage 2: Grand Finale (Mission Execution)

Objective

To demonstrate a **fully autonomous underwater system** capable of:

- Complex navigation
- Multi-sensor integration
- Precision manipulation
- Object detection & identification

Challenge Description (Upgraded)

Mission: Autonomous Target Identification & Payload Deployment

The AUV must:

- Navigate through underwater arena
- Identify visual and acoustic targets
- Accurately deploy payload
- Return safely

Advanced Challenge Tasks

1. Qualification Gate Run (Full)

- Forward pass → U-turn → Reverse pass
- No contact allowed

2. Autonomous Zone Navigation

- Navigate to target zone using **vision-based localization**

3. Multi-Target Identification

- Identify:
  - Blue drum (visual target)
  - Acoustic pinger drum
  - QR-coded fully submerged objects (5 or more)

4. Precision Payload Deployment

- Drop payload into:
  - Blue drum (decoy)
  - Pinger drum (actual target)

5. Optional Mapping Challenge (Advanced)

- Generate **2D/3D digital map** of:
  - Arena
  - Target positions

6. Return & Surface

- Return to start zone
- Controlled surfacing

3.4 Arena Design (Enhanced)

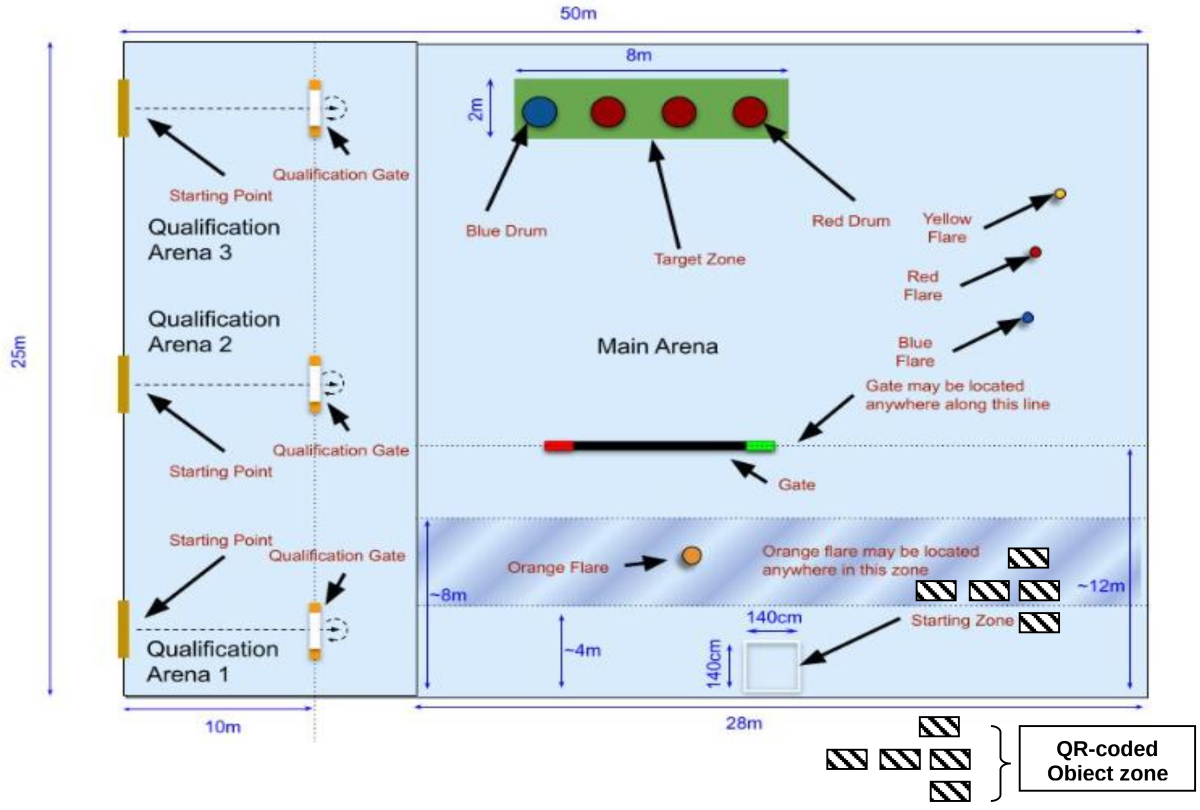
- Pool Size: 10m × 6m (minimum)
- Depth: ≥1.5m
- Dynamic placement of targets
- Optional disturbance factors:
    - Water turbulence
    - Lighting variation

3.5 Technical Requirements

- Fully autonomous operation
- Onboard processing mandatory
- Multi-sensor integration encouraged:
    - Camera
    - IMU
    - Hydrophone
    -

Evaluation Criteria (Grand Finale)

Task	Description	Points
Qualification Gate Run	Forward pass → U-turn → Reverse pass through gate without touching sides	20
Visual Target Navigation	Identify QR-coded submerged objects and Navigate to the green mat zone using camera vision	20
Target Drum Identification	Identify the blue drum and pinger-emitting red drum among 4	10
Payload Delivery	Drop the ball into:	
	- Blue Drum	30
	- Red Drum with Pinger	20
	- Any other Red Drum	10
Bonus Mapping Challenge	Post-mission map of mat layout with drum positions (2D Digital Map)	5
Return to Start Zone	Return and surface within 10-minute time limit	5
Clean Run Bonus	No surface contact or boundary hits	(10)
TOTAL MAX SCORE		100



Tie Breaker

1. Maximum tasks completed.
2. Least penalty points.
3. Minimum completion time.
4. Decision of the Technical Advisory Committee shall be final.

Penalties (-5 marks for each violation)

- Collision with gate
- Human intervention
- Robot sink
- Incorrect task execution
- Safety violation

Safety and Conduct:

- Robots must not cause damage to the field or audience.
- Judges' decision will be final in case of disputes or technical queries.

One retry is permitted at the discretion of TAC members.

Note: The guidelines related to the eligibility, timelines, formats of submissions, prizes and conduct are published on the Robofest Gujarat Webportal and shall be binding to all participating teams.